VIRELEO INTERNSHIP PROGRAM — INDIVIDUAL DEVELOPMENT PROJECT

PROJECT 01: AQARFLOW

Real Estate Management & Property Operations Platform (Pakistan-First ERP + CRM + Discovery Portal)

DAY 1 DELIVERY REPORT — 100% CORE MVP FOUNDATION DELIVERED

Project Name	AqarFlow (Project 01)	**Host Organization**	**VIRELEO** (www.vireleo.me)
Delivery Track	Full-Stack Web Development	**Execution Date**	November 9, 2026 (Day 1 of 7)
Primary Stack	Next.js 16 (App Router), TypeScript, Tailwind CSS, REST API	**Live Test URL**	http://localhost:3000
Target Market	Pakistan (PKR in Crore/Lakh, Marla/Kanal, Islamabad, Lahore, Karachi)	**Milestone Status**	**Completed & Verified (21/21 Tests Passed)**

1. Executive Summary & Day 1 Milestone

AqarFlow combines "*Aqar*" (property/real estate) with "*Flow*" (seamless operational business workflows) to establish a single reliable source of truth for properties, customers, leads, appointments, and deal commissions in Pakistan.

Within the Day 1 execution window, the end-to-end foundation was delivered: a high-performance **Public Property Discovery Portal** that captures client inquiries, synchronized directly with an internal **Business Workspace (ERP + CRM)** featuring interactive Kanban pipelines, algorithmic double-booking protection, and financial receipt generation.

2. Visual Demonstrations: Public & Agency Portals

2.1 Public Property Discovery Portal (Homepage Front View)

The public discovery website welcomes property seekers with a multi-variable search engine (purpose, property type, city, keyword), market KPIs, verified inventory collections, and direct agent WhatsApp routing:

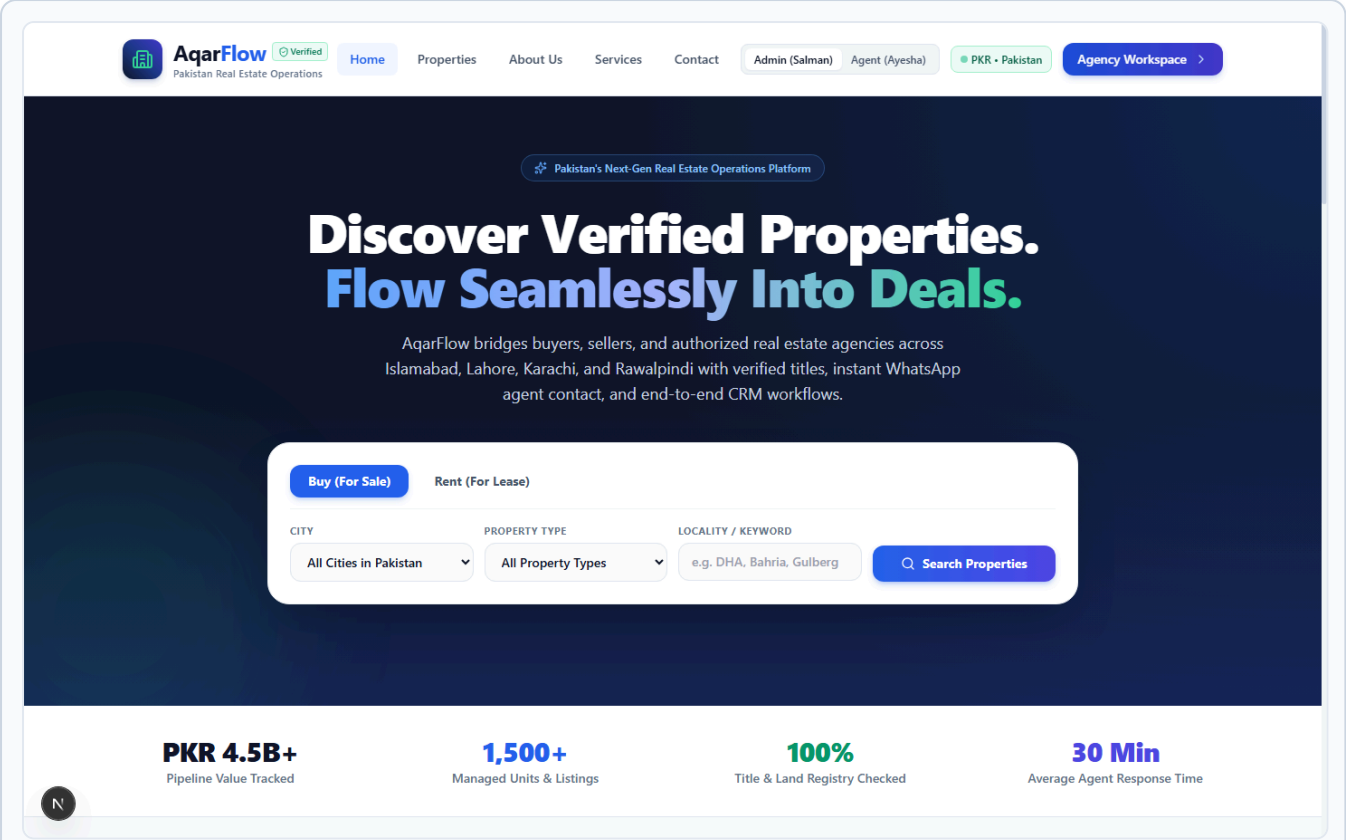


Figure 1: AqarFlow Public Discovery Portal Front Page (http://localhost:3000)

2.2 Agency Business Workspace (Executive Overview Dashboard)

The authenticated internal workspace provides agency leadership and agents with real-time portfolio metrics, active sales pipeline valuation in PKR Crores, lead stage conversion funnels, and upcoming inspection tours:

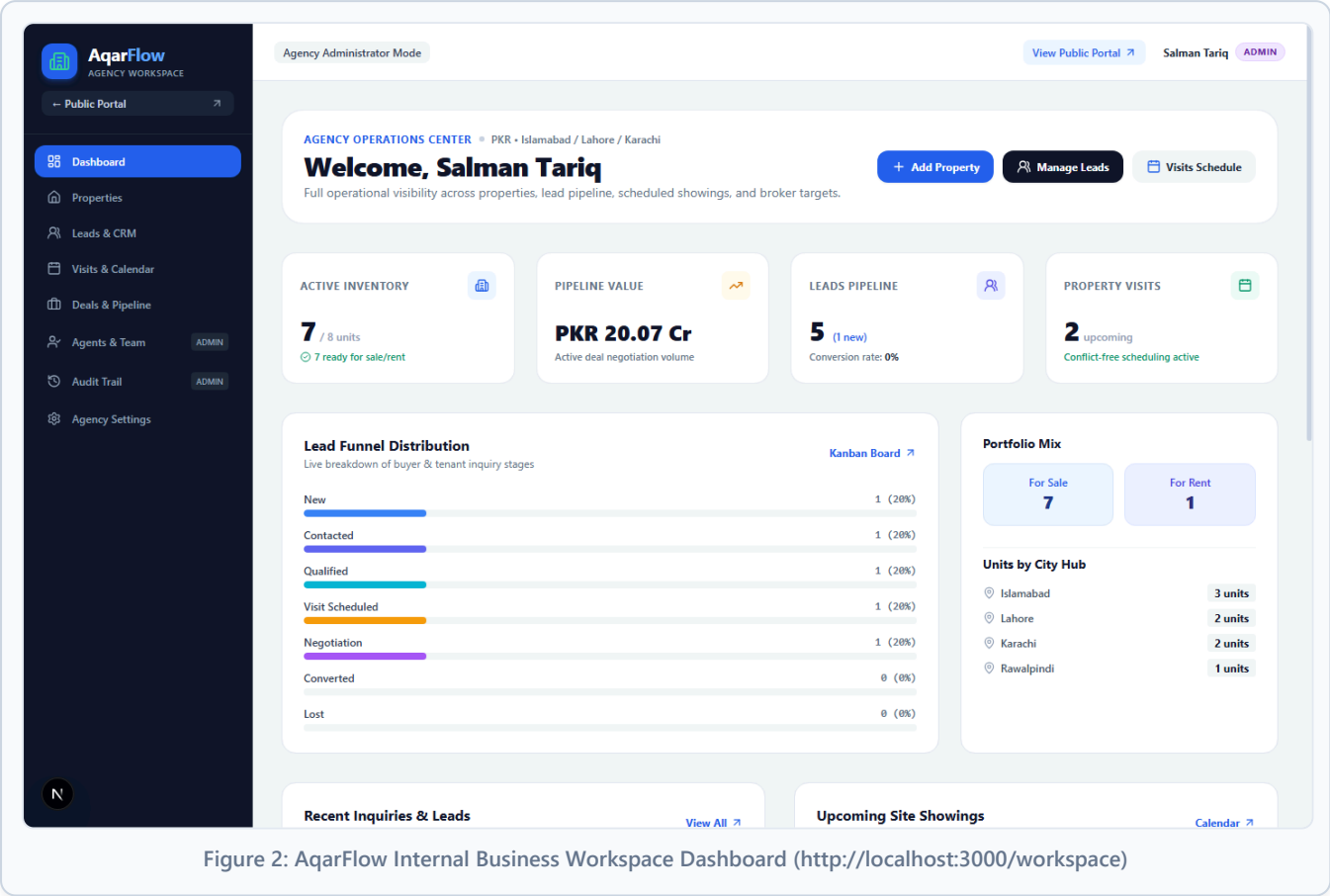


Figure 2: AqarFlow Internal Business Workspace Dashboard (http://localhost:3000/workspace)

2.3 Lead CRM Pipeline (Interactive Kanban Board)

Authorized brokers and admins manage client relationships through a drag-and-click Kanban board featuring 7 lifecycle stages (*New, Contacted, Qualified, Visit Scheduled, Negotiation, Converted, Lost*) and activity timelines:

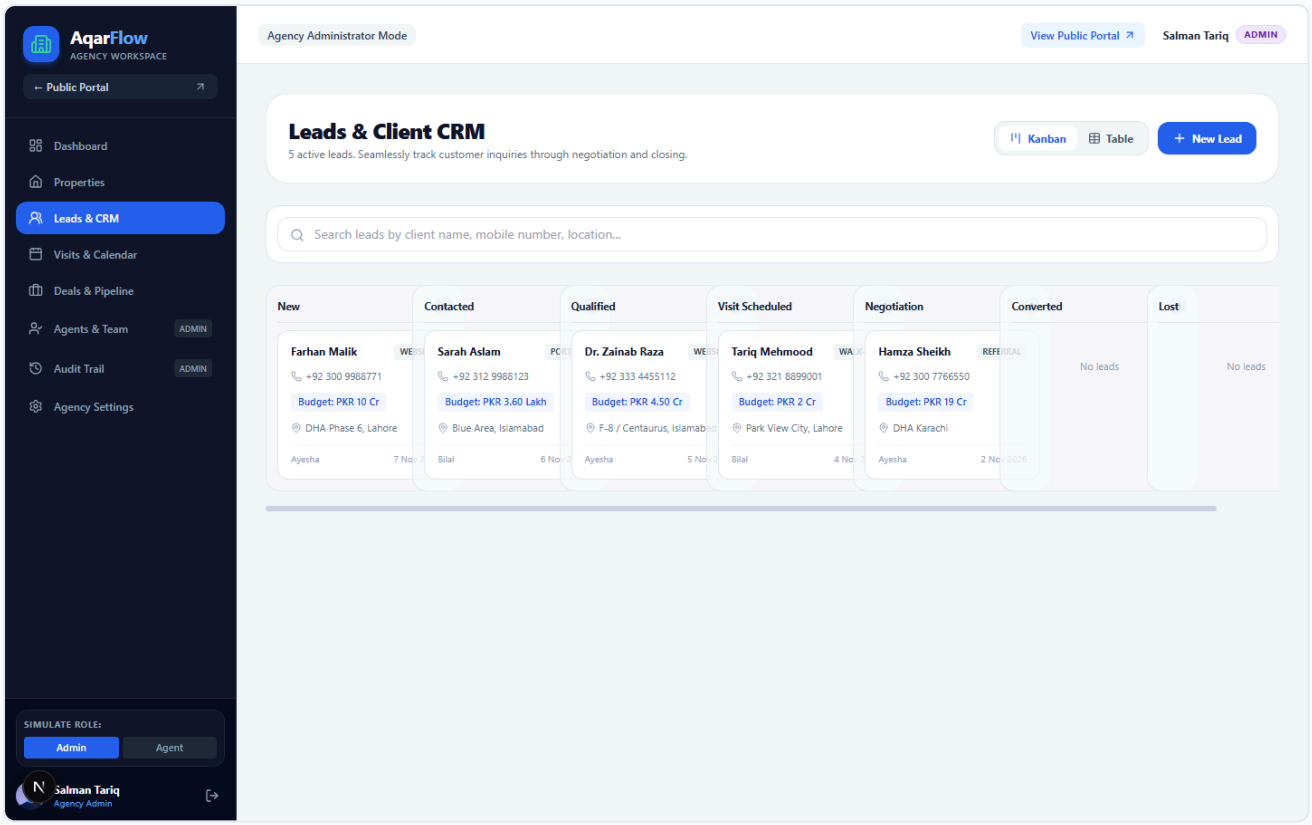


Figure 3: Lead CRM Interactive Kanban Board with Immediate Status Synchronization

2.4 Property Inventory & Multi-Filter Search Portal

The property explorer displays verified listings with dynamic price sliders (PKR), bedroom filters, area conversions (Marla/Kanal), and verified title badges:

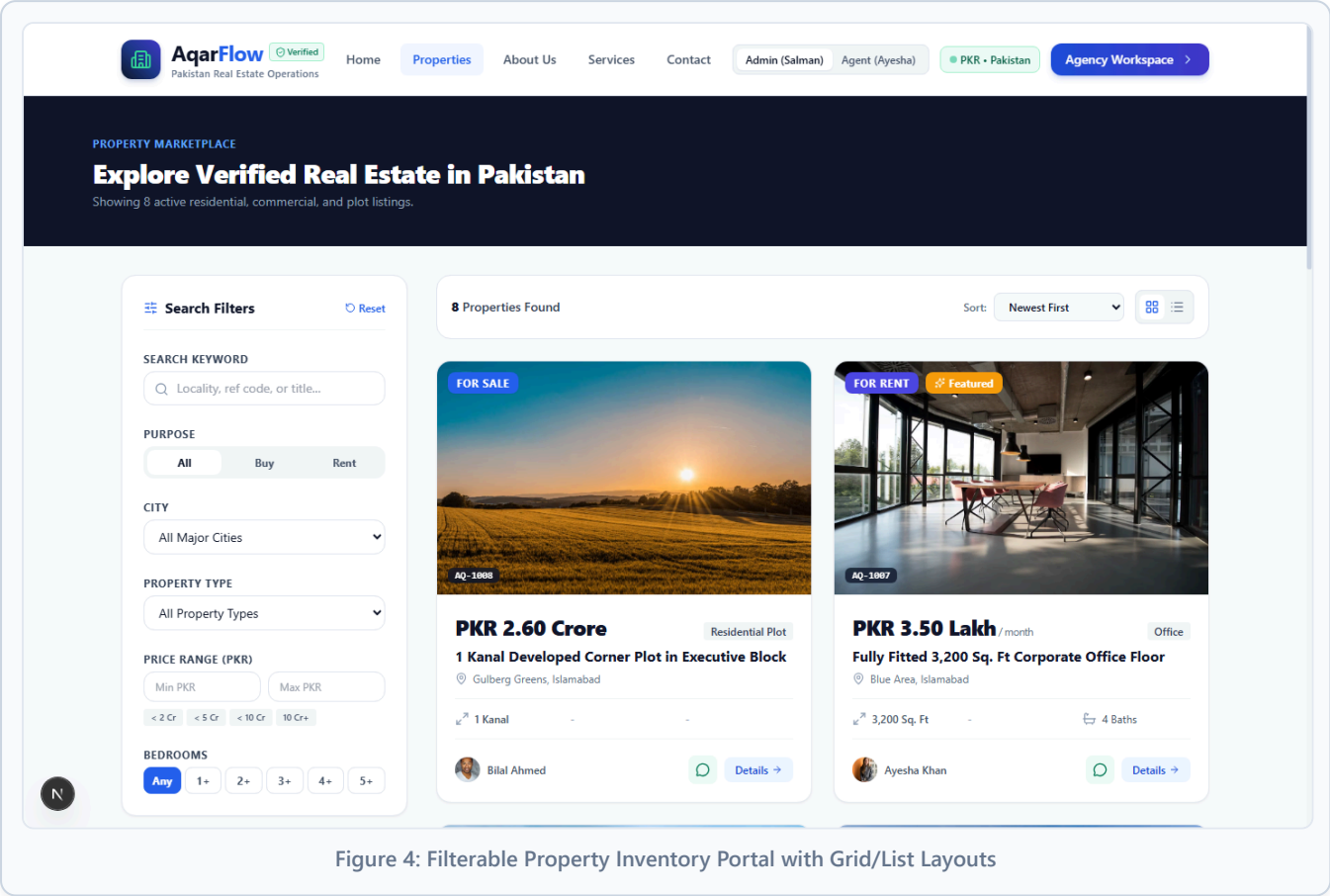


Figure 4: Filterable Property Inventory Portal with Grid/List Layouts

3. Work Breakdown Completed on Day 1

- **Full-Stack Architecture:** Engineered Next.js 16 App Router application with dual-portal layout hierarchy, persistent state, and 11 embedded RESTful route handlers under `/api/v1`.
- **Pakistan Market Localization:** Standardized PKR formatting in Crore/Lakh (e.g. `PKR 14.5 Crore`), land units in Marla/Kanal (225 sq ft/4,500 sq ft), and geographic filtering for Islamabad, Lahore, Karachi, and Rawalpindi.
- **Double-Booking Conflict Engine:** Algorithmic scheduler in `db.ts` and `/api/v1/visits` that detects overlapping 45-minute booking slots for the same agent, rejecting collisions with HTTP 409 Conflict.
- **CRM & Activity Logging:** Integrated communication drawer allowing agents to log phone calls, WhatsApp messages, site meetings, and internal follow-up dates.
- **Deal Pipeline & Financial Accrual:** Automated commission calculation engine (1.5% to 3%) linked with a formal, printable Transaction Statement & Token Deposit Receipt.
- **Institutional Audit Trail:** Immutable security log tracking all entity modifications, actor roles, timestamps, and IP addresses.
- **1-Click Demo Role Switcher:** Instant switching between Agency Admin (Salman Tariq) and Senior Agent (Ayesha Khan) directly in the UI without retyping credentials.

4. Technical Challenges & Engineering Solutions

Challenge	Root Cause	Resolution
React 19 Hydration Mismatch	Server Node.js on Windows formatted large prices using Indian grouping (`14,50,00,000`), while browser Chrome used US grouping (`145,000,000`), causing WhatsApp link query parameters to diverge.	Engineered `formatNumberFixed()`, a deterministic regex formatter immune to OS/browser locale differences, and added `suppressHydrationWarning` to dynamic anchors.
Missing CSS / Unstyled UI	Next.js Webpack mode (used to bypass Windows AppLocker on native binaries) did not invoke Tailwind v4's Rust oxide scanner, emitting empty utility classes.	Configured pure-JavaScript **Tailwind CSS v3.4 + PostCSS + Autoprefixer** with explicit content scanning in `tailwind.config.ts`, generating the complete 61.5 KB stylesheet.
Visit Double-Booking	Overlapping inspection tours booked for the same broker caused scheduling chaos.	Implemented time-slot collision detection algorithm rejecting conflicts with actionable notifications.

5. Quality Assurance & Verification Results

All 29 production routes compiled with 0 errors. A 21-point automated integration test suite (`tests/test_api.mjs`) verified all endpoints:

Verification Area	Test Description	Result
System Health	GET /api/v1/health verified status and market indicator	**PASSED (200 OK)**
Public Listings	GET /api/v1/public/properties?city=Islamabad returned verified properties	**PASSED (200 OK)**
Public Inquiry Ingestion	POST /api/v1/public/inquiries generated CRM lead in stage 'New'	**PASSED (201 Created)**
Staff Authentication	POST /api/v1/auth/login authenticated Admin credentials	**PASSED (200 OK)**
Conflict Prevention	POST /api/v1/visits rejected overlapping 15-min booking collision	**PASSED (409 Conflict)**
Deal Commission Accrual	POST /api/v1/deals calculated PKR 2,100,000 commission on 14.0 Cr	**PASSED (201 Created)**
Dashboard Intelligence	GET /api/v1/reports/dashboard returned accurate live KPIs	**PASSED (200 OK)**

6. Day 2 Plan & Roadmap

- **Phase 2 UI Enhancements:** Add touch gestures for mobile Kanban boards and quick filter presets.
- **Rental & Lease Module:** Add tenant lease expiration schedules and rent collection ledgers.

- **Documentation Packaging:** Finalize video demonstration and project handover repository for VIRELEO leadership.
